a Privileged learning

Training inputs

"... noodles as food."

Audio

Video

Frozen TAV teacher

Private encoders
TA / TV / AV
Bridge + head

Frozen targets

z_C

F

y Task labels

Trainable text-assisted branch

Masked text

AV states (b)

TA / TV
AV + head

r_T

f_{aux}

$\hat{y}$ (b)

b SpiECo: audiovisual student

Inputs **Features** **Encoder** **Residual PLIF (c)**

Audio

A

Projection
BiLSTM

H_A

$\tilde{H}_A$

θ_0

c_{AV} (d)

Video

V

Projection
BiLSTM

H_V

$\tilde{H}_V$

Masked pooling

Concatenation

Feature flow

Supervision

Initialization

Bridge

z_θ

Head

$\hat{y}$

c Residual spiking evidence dynamics

$k = 1, ..., K; K \leq 8$

PLIF

H

Temporal
binning

$\bar{h}_1, ..., \bar{h}_K$

Current
projection

I_k

$v_0 = 0$

v_k

s_k

$W_{out}s_{b(t)} + b_{out}$

$\alpha = 0.1$

$\tilde{H}$

H

d Reciprocal AV interaction

$\tilde{H}_A$

$\tilde{H}_V$

Q_A, K_V, V_V
Cross-attention

Q_V, K_A, V_A
Cross-attention

g

Gated fusion

$\tilde{H}_A$

$\tilde{H}_V$

$\frac{1}{2}\Sigma$

LN + FFN
c_{AV}